Calendar
Combinatorics
MATH 31003-01 - Fall 2026

I will put all deadlines on the Blackboard Calendar. Please check it regularly.

0.1 Key university dates

For official University dates, see the Fall 2026 academic calendar and the Registrar's final-exam schedule.

Date	Event
Monday, August 17	Classes begin
Monday, September 7	Labor Day holiday
Tuesday, November 24–Friday, November 27	Thanksgiving holiday
Wednesday, December 9	Final exam: 8:00-10:00 a.m., Wednesday, December 9, 2026

0.2 Exams

Date	Assessment
Friday, September 18 **TENTATIVE**	Midterm 1
Friday, October 16 **TENTATIVE**	Midterm 2
Friday, November 13 **TENTATIVE**	Midterm 3
8:00-10:00 a.m., Wednesday, December 9, 2026	Exam 4 and mastery reassessment

0.3 Daily lecture log

The public daily log below contains the lectures currently released. Later planning entries remain in the instructor's private lecture log and do not appear in this navigation.

	Wk	Lec
		L1: Set notation and basic counting
2026-08-17	1	Sets, cardinality, Cartesian products, and the sum rule.
		L2: Product rule and strings
2026-08-19		Product rule; strings with and without repetition.
		L3: More counting techniques
2026-08-21		Subtraction, division, permutations, combinations, and counting models.
		L4: Applications of combinations and combinations with repetition
2026-08-24	2	Lattice paths. Repeated-letter anagrams and multinomial coefficients. Combinations with r
		L5: Review of choice models, increating sequences, and basic binomial expansio
2026-08-26		Choosing the correct model: does order matter and are repetitions allowed. A second biject
		L6: Polynomials, paths, Pascal recursion and restricted lattice paths, intro to in
2026-08-28		Multinomial formula and polynomial expansions Coefficient extraction from powers of mult
		L7: Catalan paths and inclusion exclusion principle.
2026-08-31	3	Reflection argument and Catalan numbers. Inclusion–exclusion for two, three, and finitely r
		L8: Applications of inclusion–exclusion
2026-09-02 **TENTATIVE**		Conclusion of proof of inclusion exclusion principle. Complement and violation-set formulat

TENTATIVE Lecture topics and assessment dates may change.

0.4 Week and unit overview

Weeks	Unit	Topic
1–4 TENTATIVE	Unit I: Enumeration	Counting models, bijections, double counting, and incl
4–5 TENTATIVE	Unit II: Generating-function foundations	Formal power series and ordinary generating-function
6–8 TENTATIVE	Unit III: Recurrences	State models, linear recurrences, recursive evaluation,
9–16 TENTATIVE	Further topics	To be announced